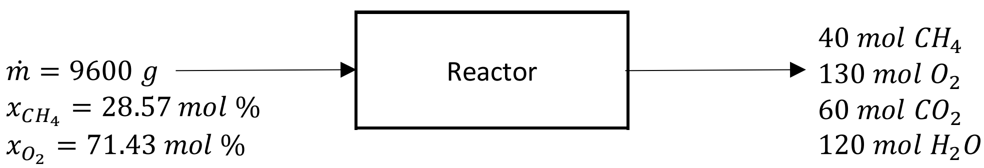
CHE210: ***Discussion Session 14*** **Name:**____________________________________

Use the information below to answer questions (1) through (3). Methane is burned to form carbon dioxide and water in a batcha reactor:

$$CH_4 + 2\,O_2 \rightarrow CO_2 + 2\,H_2O$$

The feed to the reactor and the products obtained are shown in the following flowchart:



1) What is the fractional conversion of methane?
   a. 0.9958
   b. 0.9854
   c. 0.7998
   d. 0.5998
2) What is the extent of the reaction for the reaction equation above?
   a. -120 mol
   b. -60 mol
   c. 60 mol
   d. 120 mol
3) It was decided that air is to be used to supply oxygen in the above batch reactor; but it was found that an excess of 30% air is also needed to prevent side reactions. Assume the same amount of methane is fed to the reactor and air composition of 21 mol% O_2 / 79 mol% N_2. Determine the mass of air required to meet this new criterion.
   a. 17.9 kg
   b. 35.7 kg
   c. 44.7 kg
   d. 74.3 kg
4) Suppose the formula $\Delta\hat{H} = \int_{T_1}^{T_2} C_p(T)\,dt$ is used to calculate the specific enthalpy change for a change in temperature ***and pressure*** undergone by: an ideal gas, a highly non-ideal gas, and a liquid. Rank these for which the formula will estimate the specific enthalpy most accurately.
   a. ideal gas > liquid > highly non-ideal gas
   b. liquid > ideal gas > highly non-ideal gas
   c. ideal gas > highly non-ideal gas > liquid
   d. ideal gas = liquid > highly non-ideal gas

5) The standard heat of the reaction

$$2\,CO \rightarrow 2\,C + O_2$$

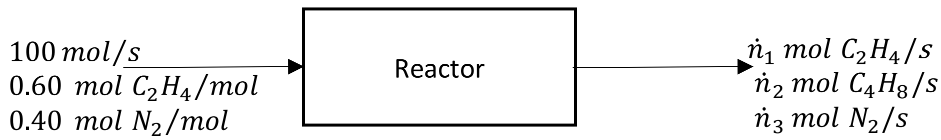
is $\Delta \hat{H}_r^\circ = +221.0 \frac{kJ}{mol}$. What is the standard heat of formation of CO?

a. -221.0 kJ/mol
b. -110.5 kJ/mol
c. 110.5 kJ/mol
d. Not enough information.

6) If substance A and substance B each have a specific gravity of 1.34, must 3 cm^3 of A have the same mass as 3 cm^3 of B?

a. Yes
b. No

7) A mixture of ethylene and nitrogen is fed to a reactor in which some of the ethylene is dimerized to butene.



How many independent atomic species balances are in the process above?

a. 0
b. 1
c. 2
d. 3

Use the information below to answer questions (8) through (10). Consider the following pair of reactions:

$$A \rightarrow 2B\ (desired)$$
$$A \rightarrow C\ (undesired)$$

Suppose 100 mol of A is fed to a batch reactor and the final product contains 10 mol of A, 160 mol of B, and 10 mol of C.

8) What is the percentage yield of B?

a. 20%
b. 40%
c. 60%
d. 80%

9) What is the selectivity of B relative to C?

a. 16
b. 8
c. 4
d. 0.0625

10) What are the extents of the first (desired) and second (undesired) reactions?

a. $\xi_1 = 160\ mol, \xi_2 = 10\ mol$
b. $\xi_1 = -80\ mol, \xi_2 = -10\ mol$
c. $\xi_1 = 80\ mol, \xi_2 = 10\ mol$
d. $\xi_1 = -160\ mol, \xi_2 = -10\ mol$

Homework Problem 6.48 (Murphy)

A waste gas stream contains 40 mol% ethane (C_2H_6), 19 mol% N_2, 1 mol% H_2S, 10 mol% O_2, and 30 mol% CO. The gas is fed to a furnace along with 20% excess air and combusted. The waste stream entering the furnace is at 25°C. The air enters the furnace at 25°C. The flue gas exits the furnace stack at 150°C. Complete combustion can be assumed. The heat generated in the furnace is used to generate saturated steam at 10 bar, using boiler feedwater at 50°C and 10 bar. Calculate the amount of steam generated in the furnace per gmol waste gas fed (kg steam per gmol gas).